

Nawa Dahal

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Education

San José State University

B.S. Applied and Computational Mathematics

GPA: 3.93/4.0

San José, California

Aug. 2025 – Expected May 2027

De Anza College

Associate of Science - AS, Mathematics

GPA: 3.78/4.00

Cupertino, California

Sep. 2022 – Jun. 2025

Research Interests

Numerical Analysis, Scientific Computing, Nonlinear Partial Differential Equations, Applied Analysis, Dynamical Systems, and Mathematical Modeling.

Research Experience

A Matrix-Based Perspective on Multi-Level Richardson Extrapolation

Undergraduate Researcher | Advisor: Dr. Mohammad Saleem

San José State University

Sep. 2025 – Dec. 2025

- Formulated multi-level Richardson extrapolation through a Vandermonde-type linear system for the extrapolation weights.
- Identified the weights as Lagrange interpolation basis polynomials evaluated at zero, connecting extrapolation with polynomial interpolation.
- Analyzed conditioning, extrapolation-weight growth, and floating-point error using MATLAB experiments with smooth, insufficiently smooth, and highly oscillatory integrands.

Analysis and Numerical Investigation of Nonlinear Semiconductor Drift-Diffusion Systems

Honors Undergraduate Researcher | Advisor: Dr. Daniel Brinkman

San José State University

Jan. 2026 – Jul. 2026

- Derived second-order finite-difference discretizations of the equilibrium Poisson-Boltzmann equation and steady-state drift-diffusion-Poisson system.
- Analyzed fixed-point and quasi-linearized iterations using contraction arguments and spectral properties of the discrete Laplacian; proved that the quasi-linearized update is equivalent to Newton's method.
- Developed a depletion-aware initial approximation and implemented Gummel iteration with Scharfetter-Gummel fluxes, Shockley-Read-Hall recombination, and Ohmic contact conditions.
- Studied convergence and current conservation under varying doping and applied bias, and computed current-voltage characteristics consistent with the Shockley diode model.

Photovoltaic Extension of a Drift-Diffusion-Poisson Solver

Independent Researcher

San José, California

Summer 2026

- Extended the one-dimensional drift-diffusion-Poisson solver from dark diode transport to illuminated solar-cell simulation by incorporating Beer-Lambert optical generation into the carrier-continuity equations.
- Computed illuminated current-voltage characteristics through terminal-voltage sweeps and extracted the short-circuit current density, open-circuit voltage, maximum-power point, and fill factor.

Two-Dimensional Semiconductor and PNP Transistor Modeling

Undergraduate Researcher | Advisor: Dr. Daniel Brinkman

San José State University

Aug. 2026 – Present

- Extended the steady-state drift-diffusion-Poisson framework from one to two spatial dimensions using finite differences, sparse linear systems, and mixed Ohmic-contact and insulating boundary conditions.
- Developed a planar PNP transistor simulation with separate emitter, base, and collector contacts, using physics-informed initialization and applied-voltage continuation to obtain convergence and examine terminal-current balance.
- Next stage: investigate a differential-algebraic formulation of the coupled system as an alternative to the linearized Poisson subproblem within Gummel iteration.

Teaching Experience

MATH 19: Precalculus Instructional Student Assistant: Workshop Facilitator	San José State University Aug. 2025 – Dec. 2025
- Independently led weekly workshop sessions for approximately 25 students, guiding collaborative problem solving and mathematical discussion. - Administered and graded student assessments and provided feedback on students' mathematical reasoning.	
MATH 33LA: Differential Equations and Linear Algebra Instructional Student Assistant: Grader	San José State University Jan. 2026 – May 2026
- Graded quizzes, midterms, and the final examination for a large-enrollment course of approximately 160 students. - Evaluated students' solutions using detailed rubrics, identified conceptual and computational errors, and provided written feedback.	
MATH 71W: Calculus Workshop for Business and Aviation Instructional Student Assistant: Workshop Facilitator	San José State University Aug. 2026 – Present
- Lead weekly workshop sessions and guide students through collaborative problem solving in applied calculus. - Administer and grade student assessments and reinforce students' understanding through individual and group instruction.	
MATH 143C: Numerical Analysis and Scientific Computing Instructional Student Assistant: Grader	San José State University Aug. 2026 – Present
- Grade homework and computational projects for an upper-division numerical analysis course. - Evaluate mathematical derivations, numerical algorithms, and computational results for accuracy and clarity.	

Publications

Dahal, N. (2026). A matrix-based perspective on multi-level Richardson extrapolation. *SIAM Undergraduate Research Online*, 19, 409–428. <https://doi.org/10.1137/26S1857207>

Honors Thesis

Analysis and Numerical Investigation of Nonlinear Semiconductor Drift–Diffusion Systems	2026
Undergraduate honors thesis, Department of Mathematics and Statistics, San José State University. Advisor: Dr. Daniel Brinkman.	

Posters & Presentations

Numerical Investigation of the Poisson–Boltzmann Equation Research Poster	San José State University Apr. – May 2026
- Presented at the College of Science Student Research Day (Apr. 2026). - Presented at the Department of Mathematics and Statistics End-of-Year Research Showcase (May 2026).	

Honors & Awards

2022–2025 Dean's List (Every Quarter Enrolled) , De Anza College	Cupertino, California
Fall 2025 President's Scholar , San José State University	San José, California

Relevant Coursework

In Progress	Partial Differential Equations, Complex Analysis, Advanced Dynamical Systems
Completed	Introduction to Analysis, Advanced Calculus, Ordinary Differential Equations and Dynamical Systems, Numerical Analysis I, Numerical Analysis II, Applied Probability and Statistics

Planned Coursework

Applied Mathematics I (Graduate Partial Differential Equations), Numerical Solution of Partial Differential Equations, Mathematical Modeling for Life Sciences

Technical Skills

MATLAB | C++ | LaTeX